1. Radio, microwave, infrared, visible, UV, X-ray and gamma are all the same phenomenon — only λ is different. Use $c = \lambda f$.

A. A Wi-Fi router broadcasts at $f = 5.0$ GHz. What is its wavelength? Which band is that?

B. A wave has $\lambda = 600$ nm. What is its frequency? Which band is that?

C. Rank **FM radio**, **green light**, **X-ray** and **microwave** by increasing wavelength.

shortest $\longrightarrow$ longest

_____ _____ _____ _____

2. A bare 150 W light bulb hangs in the middle of a room and radiates uniformly in all directions.

A. What is the intensity of its light 2.0 m away?

B. Without starting over, what is the intensity 6.0 m away? Explain how you got it.

C. What is the peak electric field E_0 at 2.0 m? $\left(I = \frac{1}{2}\epsilon_0 c E_0^2\right)$

D. What is the peak magnetic field B_0 there?

3. A laser delivers an intensity of $I = 5000$ W/m^2 onto a small square of **black** card of area $A = 0.020$ m^2. The card absorbs all of the light.

A. What is the radiation pressure on the card?

B. What force does the light exert on it?

C. The card is swapped for a **mirror** of the same size. Now what is the force? Why is it different?

D. The mirror has a mass of 0.50 g and floats freely in space. What is its acceleration?